

## Scalability & Future Plan

|               |                                   |
|---------------|-----------------------------------|
| Date          | 15 July 2026                      |
| Team ID       |                                   |
| Project Name  | Personalized Networking Assistant |
| Maximum Marks | 1 Mark                            |

### Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

### Current System Limitations:

| S.No | Limitation                        | Impact                         | Priority to Address (High / Medium / Low) |
|------|-----------------------------------|--------------------------------|-------------------------------------------|
| 1    | Supports limited concurrent users | May slow down with many users  | High                                      |
| 2    | Stores history in JSON files      | Difficult to manage large data | Medium                                    |
| 3    | Basic authentication and security | Less secure for production     | High                                      |

### Scalability Plan:

| S.No | Scalability Aspect | Current State                    | Proposed Upgrade / Solution  |
|------|--------------------|----------------------------------|------------------------------|
| 1    | User Load          | Supports a small number of users | Deploy on cloud servers      |
| 2    | Data Storage       | Uses JSON files                  | Use PostgreSQL or MongoDB    |
| 3    | Performance        | Runs on a local machine          | Add caching and optimization |
| 4    | Security           | Basic security                   | Add login and authentication |

### Future Roadmap:

| Phase | Planned Feature / Enhancement | Target Timeline | Expected Impact |
|-------|-------------------------------|-----------------|-----------------|
|-------|-------------------------------|-----------------|-----------------|

|         |                            |          |                                 |
|---------|----------------------------|----------|---------------------------------|
| Phase 1 | User authentication system | 1 month  | Secure access                   |
| Phase 2 | Multi-language support     | 2 months | More Users                      |
| Phase 3 | Cloud deployment           | 3 months | Better scalability              |
| Phase 4 | AI model improvement       | 4 months | Better conversation suggestions |